UFR de mathématique et d'informatique
Université de Strasbourg

MASTER D'INFORMATIQUE
PARCOURS SDSC
SCIENCES DES DONNÉES ET SYSTÈMES COMPLEXES

Travail d'Étude et de Recherche

Kian ABDOLMOHAMMADIAN
kian.abdolmohammadian@etu.unistra.fr

IMPROVING SEGMENT ANYTHING MODEL ON REMOTE SENSING IMAGERY

May 19, 2026

Research work supervised by

Baptiste LAFABREGUE
lafabregue@unistra.fr

Abstract

The Segment Anything Model (SAM) [10] offers strong zero-shot segmentation capability but was designed for natural ground-level images: applying it to large aerial remote sensing images [8, 13] raises three challenges, fixed input resolution requiring tiling, class-agnostic output requiring classification, and boundary fragmentation requiring post-hoc merging.

This work studies and evaluates a complete processing pipeline addressing all three challenges on the ISPRS Potsdam benchmark [15] (19 tiles, 6 classes, 5 cm/pixel). On the tiling side, overlapping sliding-window inference [1] with Non-Maximum Suppression is compared to a fixed non-overlapping grid. On the merging side, geometry-based methods (RAG [6], UCM [2]) are applied and evaluated against class-guided strategies (SMM [11], raster gap filling). For segment classification, four feature representations of increasing richness are evaluated: mean RGB K-means, DINOv2 [12] CLS embeddings (64 px crop), DINOv2 CLS combined with RGB, and multiscale DINOv2 with overclustering (k=24) [7], alongside supervised Random Forest and MLP baselines.

The central finding is that the order of operations matters: classifying segments first, then merging per class, is the correct design. Geometry-only merging reduces fragmentation but inevitably introduces cross-class fusions; class-guided merging avoids this at the cost of requiring predicted labels. DINOv2 multiscale embeddings reliably break the RGB spectral bottleneck, enabling effective unsupervised classification, and overclustering further improves alignment accuracy at the cost of cluster interpretability.

Keywords: remote sensing, image segmentation, Segment Anything Model, tiling, DINOv2, K-means, overclustering, mask merging, ISPRS Potsdam.

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Chapter 1

Introduction

1.1 General context

Image segmentation consists in dividing an image into regions that correspond to distinct objects or areas. In remote sensing, this task is essential for the automatic analysis of large geographic areas from aerial or satellite imagery. Concrete applications range from urban planning, mapping building footprints, road networks, and land-use zones from city-scale orthophotos, to precision agriculture, where delineating field parcels and vegetation patches supports crop monitoring and yield estimation. Environmental agencies use segmentation maps for deforestation detection and flood extent mapping from multi-temporal satellite imagery, tasks where timely and automated processing of large areas is critical. In defence and infrastructure contexts, detecting vehicles, runways, or construction sites at scale is likewise only feasible if the segmentation step can be fully automated.

Segmentation in this domain remains difficult for several reasons. Images cover large areas at high resolution, objects vary widely in size and shape depending on the sensor and the scene, and producing the pixel-level annotations required to train supervised models is expensive and time-consuming. These practical constraints make zero-shot or weakly supervised approaches particularly attractive: a system that can segment and label new imagery with little or no manual annotation could dramatically reduce the cost of deploying remote sensing analysis at scale.

1.2 The Segment Anything Model and remote sensing

The Segment Anything Model (SAM) [10], introduced by Kirillov et al. in 2023, is a foundation model capable of generating object masks from simple prompts such as points or bounding boxes, without any task-specific training. This zero-shot capability makes SAM attractive for remote sensing, where labelled data is scarce. However, SAM was trained on natural ground-level photographs, and its performance on overhead imagery is often limited without adaptation.

1.2.1 The tiling problem

A further constraint is that SAM's image encoder accepts a fixed input size of 1024×1024 pixels. Large remote sensing images must therefore be divided into tiles before inference. This introduces artefacts at patch borders: objects can be split across tiles, duplicate masks can appear in overlapping regions, and boundaries may be inconsistent between adjacent tiles. These

issues reduce the spatial coherence of the final segmentation and affect the reliability of downstream analyses.

1.3 Research question

This work addresses the following question: how can SAM be applied to large remote sensing images while producing spatially consistent segmentation results? More specifically, to what extent do the tiling strategy, the mask merging method, and the boundary correction technique influence the quality of the output?

1.4 Objectives and contributions

This work studies and evaluates a complete pipeline for applying SAM to large remote sensing images. Several existing methods are adapted and integrated — RAG and UCM graph-based merging [6, 2], SMM correlation clustering [11], and SAHI-style overlapping tiling [1] — and evaluated in this remote sensing context for the first time as part of a unified pipeline.

The personal contributions of this work are:

- **RGB K-means baseline.** A K-means clustering directly on mean RGB segment statistics, serving as the spectral lower bound for classification.
- **DINOv2 segment descriptors.** Extraction of DINOv2 ViT-B/14 [12] CLS embeddings from 64 px crops centred on each SAM segment, demonstrating that a tight crop outperforms a wider context crop.
- **DINOv2 + RGB fusion.** Concatenation of DINOv2 CLS embeddings with mean RGB statistics, evaluated across three crop scales (64, 128, 256 px) into a 2 304-dimensional multiscale descriptor.
- **Overclustering strategy.** Application of K-means with k=24 on the multiscale DINOv2 + RGB descriptor, combined with greedy many-to-one alignment, showing that overclustering captures intra-class appearance variability and substantially raises post-alignment accuracy.

Chapter 2

Related work

2.1 Segment Anything Model

SAM [10] is a segmentation foundation model introduced by Meta AI in 2023. It is composed of three main components: an image encoder, a prompt encoder, and a mask decoder.

The image encoder is based on a Vision Transformer (ViT) pre-trained with masked autoencoders. It takes a 1024×1024 pixel image as input and produces a dense feature embedding. The prompt encoder handles sparse prompts such as points and bounding boxes, as well as dense prompts such as rough masks. These representations are combined in a lightweight mask decoder that predicts one or several candidate masks along with confidence scores.

SAM was trained on SA-1B [10], a dataset of over one billion masks built through a data engine combining automatic annotation and human correction. This gives SAM strong zero-shot generalisation: it can segment objects in new images without task-specific fine-tuning.

Despite these strengths, SAM has known limitations. Its fixed input resolution constrains how large images can be processed. Its training data consists almost exclusively of natural ground-level photographs, which differ substantially from overhead imagery. Finally, SAM produces class-agnostic masks: it segments objects but does not assign semantic labels to them.

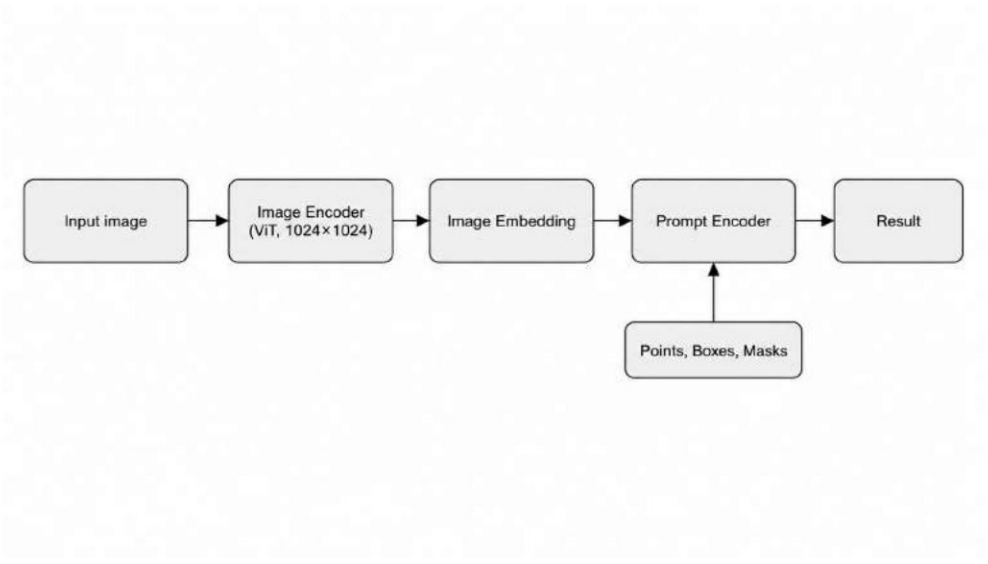


Figure 2.1: Simplified overview of the SAM architecture (based on [10]).

2.2 SAM applied to remote sensing

Several evaluations have shown that SAM's zero-shot performance on aerial and satellite imagery is significantly lower than on natural images [8, 13, 19]. Objects such as buildings, vehicles, or agricultural fields are often poorly detected without carefully chosen prompts, and the model tends to produce fragmented or over-segmented results due to the domain gap between its training data and overhead imagery.

To address this, several adaptation strategies have been proposed in the literature. A first approach consists in adapter-based fine-tuning: SAM's image encoder is frozen and lightweight adapter modules are inserted and trained on remote sensing data. SAM-Adapter [3] and RSPrompter [4] follow this pattern, achieving improved performance at a low computational cost while preserving SAM's general capabilities. RingMo-SAM [21] extends this idea with a larger-scale fine-tuning on remote sensing datasets.

A second approach focuses on prompt engineering. Since SAM is prompt-driven, using denser or better-informed prompts can significantly improve results. Cheng et al. [5] show across multiple datasets and modalities that bounding box prompts consistently yield the best zero-shot segmentation performance compared to point or mask prompts, a finding that extends beyond the medical domain. In remote sensing, Osco et al. [13] evaluate SAM on UAV, airborne, and satellite imagery using point, bounding box, and text prompts, confirming that bounding boxes provide the most reliable results. Grid-based point sampling and tuned automatic mask generation parameters are further strategies commonly used in this direction.

SAMGeo [20] is a geospatial extension that integrates geographic metadata and projection handling, making SAM easier to use in GIS workflows without modifying the model itself. SAMRS [19] provides a large-scale benchmark for evaluating and fine-tuning SAM on remote sensing segmentation tasks.

SAM 2 [14], the successor to SAM, introduces a memory mechanism for propagating masks across video frames. While primarily designed for video, this architecture is relevant for multi-temporal remote sensing analysis.

2.3 Processing large images

Since SAM's encoder requires a fixed 1024×1024 pixel input, large images must be divided into tiles before inference. The simplest approach is a regular grid partition without overlap, but this tends to produce many boundary artefacts. A common improvement is to use overlapping tiles so that objects near a border appear fully in at least one tile.

Slicing Aided Hyper Inference (SAHI) [1] formalises this process: it slices the image into overlapping patches, runs inference on each patch, and merges the predictions into a full-image result. SAHI has been adapted for use with SAM and provides a practical baseline for large image processing. The choice of patch size and overlap ratio directly affects output quality and inference time, making it an important practical parameter to study.

2.4 Boundary correction and mask post-processing

After tiling and inference, patch results must be merged into a coherent full-image segmentation. Two main problems arise: duplicate detections and inconsistent boundaries.

Duplicate masks occur when the same object is detected in several overlapping tiles. Non-

Maximum Suppression (NMS) [1] is a standard technique that discards masks with high overlap with a higher-confidence prediction. Weighted Box Fusion (WBF) [17] is an alternative that merges overlapping predictions rather than discarding them, which can produce more accurate boundaries.

Spatial Mask Merging (SMM) [11] addresses this problem more formally by casting mask merging as a correlation clustering optimisation. A weighted spatial graph is built using R-tree indexing, with edge weights computed from spatial proximity, mask overlap, and confidence scores. Clusters are then formed by solving an integer linear program or a greedy approximation, and masks within each cluster are fused. This approach handles complex overlapping configurations more robustly than NMS alone.

At the pixel level, SegFix [22] improves boundary quality by replacing uncertain boundary predictions with values from more reliable interior regions, in a model-agnostic manner.

At a higher level, Region Adjacency Graphs (RAG) represent segments as nodes and their shared boundaries as edges. Graph-based merging methods [6] use this structure to merge similar adjacent regions, reducing over-segmentation and producing cleaner object boundaries. The Ultrametric Contour Map (UCM) [2] encodes boundary strength hierarchically, allowing segmentation at multiple levels of granularity.

2.5 Segment classification and feature representations

SAM produces class-agnostic masks, so a separate step is needed to assign semantic labels to detected segments. Different feature representations can be used for this purpose.

The simplest approach is to compute statistics from the raw RGB pixel values within each segment, such as the mean colour or colour histograms. While easy to compute, these features carry limited semantic information and are sensitive to illumination changes.

DINOv2 [12] is a self-supervised Vision Transformer that produces rich semantic features without labelled data. Its embeddings capture high-level visual concepts and generalise well across domains, making them a strong alternative to raw pixel features for segment description.

Once segments are described by feature vectors, clustering methods such as K-means or HDBSCAN can group them into semantic categories. Overclustering strategies [7] first produce a large number of clusters and then merge similar ones, improving spatial coherence and reducing sensitivity to the initial number of classes.

IIC [9] takes a different approach: it trains a network to maximise mutual information between the cluster assignments of augmented versions of the same image patch, producing spatially consistent groupings without any labels or feature backbone.

Chapter 3

Proposed Methodology

3.1 Dataset

All experiments are conducted on the ISPRS Potsdam benchmark [15], a widely used dataset for urban remote sensing segmentation. It consists of high-resolution RGB aerial images with a ground sampling distance of 5 cm per pixel, annotated at the pixel level with six semantic classes: impervious surface, building, low vegetation, tree, car, and clutter.

We work with a subset of 19 tiles, each of size 6000×6000 pixels (300 m $\times$ 300 m). These tiles are provided with pre-existing SAM segments in a vector layer called *Label_SAM_Disjoint*, which contains approximately 34 000 polygons, each labelled with its dominant ground-truth class. The image `top_potsdam_2_10_RGB` is used as the main test tile throughout the experiments, as it presents a relatively balanced distribution across all six classes.

The overall pipeline follows three sequential steps: a tiling strategy divides the image into patches, SAM segments each patch independently, and a merge strategy reassembles the results into a full-image segmentation. A separate classification step then assigns semantic labels to the class-agnostic SAM masks.



Figure 3.1: Overview of the main test tile (`top_potsdam_2_10_RGB`). (a) Original RGB aerial image. (b) *Label_SAM_Disjoint*: SAM segments with ground-truth class labels, used as the primary evaluation reference. (c) *Label_SAM_Merged*: a more fragmented representation showing segment intersections with the ground-truth boundaries.

3.2 Tiling strategies

SAM's image encoder accepts a fixed 1024×1024 pixel input, so large remote sensing images must be divided into tiles before inference. Two strategies are compared.

3.2.1 Fixed non-overlapping grid

The first strategy, referred to as *GroundBase*, divides the image into a regular 10×10 grid of non-overlapping 600×600 px tiles. SAM segments each tile independently.

The fundamental limitation of this approach is structural: an object located at a tile boundary is cut in two, and the two resulting fragments appear in different tiles with no shared pixels. Post-hoc merging cannot reliably bridge this gap, since the fragments are spatially disjoint. This produces a visible grid artefact throughout the final segmentation.

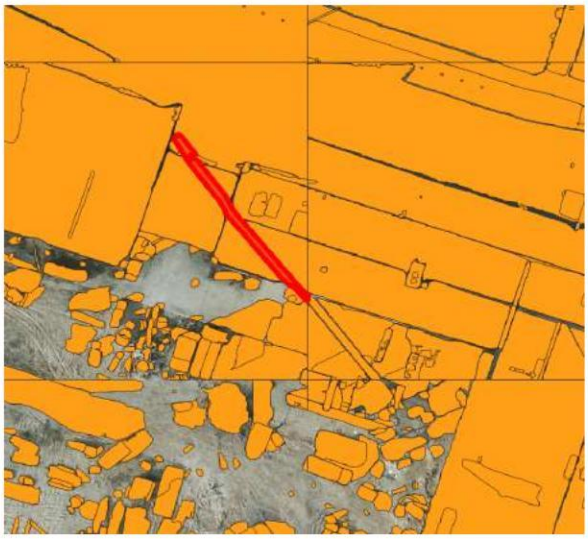
3.2.2 Overlapping sliding window

The second strategy applies overlapping tiling [1] with 600×600 px tiles and a stride of 480 px (20 % overlap). Each object near a tile boundary appears fully in at least one neighbouring tile, so SAM can produce a complete mask for it. Non-Maximum Suppression (NMS) is then applied with an IoU threshold of 0.5 to remove duplicate detections in the overlapping regions.

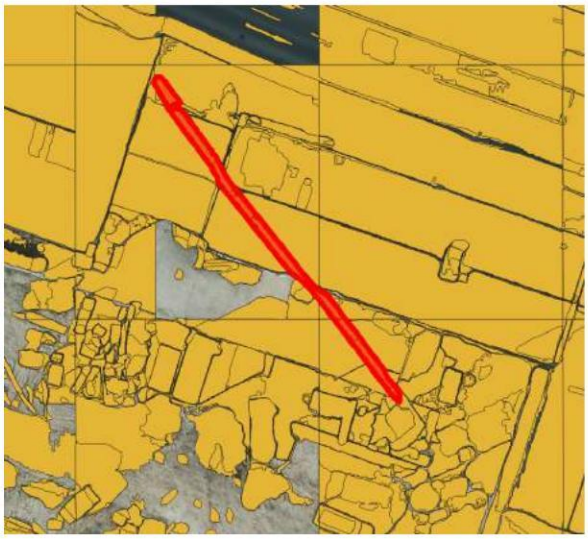
This approach partially reduces boundary fragmentation for objects smaller than the tile size (30 m): each such object appears fully in at least one tile, so SAM can generate a complete mask for it. However, as visible in Figure 3.2, tiling artefacts remain pervasive across the output, because NMS must still adjudicate between overlapping predictions and does not guarantee seamless reconstruction at every border. Objects larger than 30 m may still be split, as they cannot fit entirely within a single tile.



Figure 3.2: Comparison of GroundBase (left) and overlapping tiling (right) outputs on `top_potsdam_2_10_RGB`. Tiling artefacts are reduced but not eliminated: boundary discontinuities remain visible in both outputs.



(a) Non-overlapping tiling: the crane is split across the tile boundary.



(b) Overlapping tiling: the crane is recovered as a single complete mask.

Figure 3.3: Effect of tiling strategy on a crane straddling a tile boundary. (a) With non-overlapping tiles (GroundBase), the object is cut into two disjoint fragments that post-hoc merging cannot reliably reconnect. (b) With overlapping tiles, the crane appears fully within at least one tile, allowing SAM to produce a single coherent mask.

3.3 Geometry-based mask merging

Geometry-based methods are class-agnostic: they rely only on spatial relationships between masks, without using semantic labels.

3.3.1 Non-Maximum Suppression

When overlapping tiles are used, the same physical object can be detected independently in several patches, producing duplicate masks in the overlap zones. SAHI [1] addresses this with a post-processing NMS step applied globally after all patch predictions are pooled into a single set.

The algorithm is greedy. Let $\mathcal{P} = \{p_1, p_2, \ldots\}$ be the pool of all predicted masks sorted by SAM confidence score in decreasing order. Starting from the highest-confidence mask p_1, any subsequent mask p_j ($j > 1$) is discarded if its geometric IoU with a retained mask p_i ($i < j$) exceeds a threshold τ:

$$\text{IoU}(p_i, p_j) = \frac{\text{area}(p_i \cap p_j)}{\text{area}(p_i \cup p_j)} > \tau \;\Rightarrow\; \text{discard } p_j.$$

We use $\tau = 0.5$, the standard value recommended by SAHI [1]. The geometric intersection and union are computed directly on the polygon geometries rather than on bounding boxes, which is more accurate for the irregular shapes produced by SAM but also more expensive.

NMS has two known limitations in this setting. First, it is greedy: once a mask is retained, it suppresses all sufficiently overlapping candidates regardless of whether they correspond to the same object or to a genuinely distinct one. Second, it only removes duplicates; it does not reconnect the two halves of an object that was split at a tile boundary with no overlap between the fragments, that case requires a merging step rather than a suppression step. NMS is not

applicable to the GroundBase pipeline, since its non-overlapping tiles produce no duplicate detections.

3.3.2 Region Adjacency Graph merging

A Region Adjacency Graph (RAG) is built over the output polygons: two polygons are connected by an edge if they are spatially adjacent or within a 0.05 m buffer. The edge weight measures visual dissimilarity between the two segments:

$$w_{ij} = \frac{\|\overline{\text{rgb}}_i - \overline{\text{rgb}}_j\|_2}{\sqrt{3}} + \frac{1 - \cos(\mathbf{d}_i, \mathbf{d}_j)}{2}$$

where $\overline{\text{rgb}}$ is the mean RGB colour of a segment (normalised to $[0,1]^3$) and $\mathbf{d}$ its L2-normalised DINOv2 CLS embedding (the DINOv2 term is optional). Pairs with w_{ij} below a threshold τ are merged; no class labels are involved at any stage.

Merging uses **average linkage**: after each fusion, the centroid colour and embedding of the new component are recomputed as a weighted average of those of its members, and all outgoing edges are reweighted accordingly. This prevents the chaining artefact of single linkage, where a sequence of locally similar but globally dissimilar segments can be collapsed into one large region.

A **tile-aware** mode discounts cross-tile edges by a factor of 0.7 before merging: edges whose two segment centroids straddle a tile boundary are assigned $w_{ij} \leftarrow 0.7\, w_{ij}$, actively encouraging the fusion of fragments that were split by tiling.

The **Ultrametric Contour Map (UCM)** [2] extends this greedy strategy into a full merge hierarchy: the average-linkage procedure is run to completion while recording snapshots of the partition at thresholds $\tau \in \{0.05, 0.10, 0.15, 0.20, 0.25\}$, producing a family of segmentations at increasing levels of coarseness.

Table 3.1 shows the effect of RAG merging on the GroundBase output with conservative and aggressive parameter settings.

Both RAG and UCM produce visually smoother segmentations with fewer fragments than the baseline. However, as illustrated in Figure 3.4, they merge segments that have no semantic relationship: a road polygon adjacent to a building polygon may be fused simply because they share a long boundary. Since no class information guides the merging decision, geometrically coherent fusions are not necessarily semantically correct. This is the core limitation of agnostic merging: without labels, spatial proximity alone is insufficient to determine whether two segments belong to the same object.

Configuration	Polygons	Reduction	$<5\,\text{m}^2$	Median area
Baseline (GroundBase)	1 973	–	40.0 %	7.24 m^2
RAG conservative	1 606	18.6 %	34.9 %	8.66 m^2
RAG aggressive	1 473	25.3 %	31.2 %	9.45 m^2

Table 3.1: Effect of RAG merging on polygon count and size distribution (`top_potsdam_2_10_RGB`).



Figure 3.4: Geometry-based merging applied to `top_potsdam_2_10_RGB`. (a) Baseline Label_SAM_Disjoint (1 973 polygons). (b) RAG merging (979 polygons): adjacent segments are fused regardless of their semantic class. (c) UCM merging (683 polygons): boundary-strength-based hierarchical fusion, producing even larger regions. Both methods reduce fragmentation but introduce semantically incorrect merges, motivating the class-based approaches in the next section.

3.4 Class-based mask merging

Class-based methods assign semantic labels to segments before fusion, enabling more targeted merging decisions. Fusing two fragments of the same predicted class is a much stronger criterion than geometric proximity alone, since it ensures that the merged region corresponds to a coherent semantic object.

3.4.1 Spatial Mask Merging

SMM [11] is applied on top of the class-labelled segments. For each class, a weighted spatial graph $G = (V, E, w)$ is built using R-tree indexing: two masks i, j of the same class are connected by an edge if their centroids lie within search radius ρ. The signed edge weight is:

$$w_{ij} \;=\; \beta_1\, \phi_d(d_{ij}) + \beta_2\, \phi_{\mathrm{IoU}}(\mathrm{IoU}_{ij}) + \beta_3\, \phi_c(c_i, c_j) \;-\; \lambda$$

where $\phi_d \in [0,1]$ decreases with centroid distance d_{ij} and vanishes beyond threshold τ_d; $\phi_{\mathrm{IoU}} \in [0,1]$ is the mask overlap compatible with threshold τ_i; and $\phi_c \in [0,1]$ is the geometric mean of the two SAM confidence scores. The subtracted penalty λ makes w_{ij} negative whenever the three compatibility signals are jointly weak, so the optimiser only merges pairs that are genuinely similar.

Correlation clustering is then solved by maximising $\sum_{(i,j)\in E} w_{ij}\, x_{ij}$ over binary variables $x_{ij} \in \{0,1\}$ subject to transitivity constraints

$$x_{ij} + x_{jk} - x_{ik} \leq 1 \quad \forall\, (i,j),(j,k) \in E,$$

using an integer linear programme (ILP) or, when the ILP times out, a greedy approximation that iterates over edges in decreasing weight order and merges a pair whenever transitivity would not be violated. Masks within each resulting cluster are fused.

Since the ILP solver does not scale beyond approximately 2 500 segments per class, SMM is run independently per class. For the three smallest classes (low vegetation, tree, car), the ILP

converges and produces reductions of 49 % to 63 %. For the three dominant classes (impervious surface, building, clutter), the ILP times out and falls back to no merge. For these larger classes, the raster-based method described next provides complementary coverage.

Parameter	Role	Value
ρ	R-tree search radius (scaled px)	40
τ_d	Distance compatibility threshold	20
τ_i	IoU compatibility threshold	0.05
β_1	Distance weight	0.4
β_2	IoU weight	0.4
β_3	Confidence weight	0.2
λ	Merging penalty	1.0
γ	Anti-chaining threshold	0.5
scale	Mask rasterisation factor	0.2

Table 3.2: SMM hyperparameters used in all experiments. Distance values are in scaled pixels (scale = 0.2 maps the 6000 px image to 1200 px; $\rho = 40$ corresponds to approximately 10 m and $\tau_d = 20$ to approximately 5 m on the ground).



(a) Before: *Label_SAM_Disjoint* (b) After: SMM fusion

Figure 3.5: SMM fusion applied to a cropped region of `top_potsdam_2_10_RGB`. The target L-shaped building (highlighted in red) spans a tile boundary. (a) In *Label_SAM_Disjoint*, it is split into two separate fragments. (b) After SMM fusion, the fragments are merged into one coherent polygon.

3.4.2 Raster gap filling

To address the gap problem directly, a raster-based correction is applied in three steps.

Step 1 — Rasterisation. The vector polygons are rasterised onto the full-resolution image grid (pixel size 0.05 m) into a label map L, where $L[p] = 0$ denotes an unlabelled gap pixel.

Step 2 — Candidate identification and conflict resolution. For each class $c \in \{1, \ldots, 6\}$, a binary closing is applied to its binary mask $M_c = (L = c)$ using a disc-shaped structuring element of radius $r = \lfloor d_{\max}/0.05 \rfloor$ pixels, where $d_{\max}$ is the maximum gap parameter. The closed mask $\tilde{M}_c$ identifies void pixels that fall within distance $d_{\max}$ of class c. Since two classes

may both claim the same void pixel, conflicts are resolved using the Euclidean Distance Transform (EDT) [16]: for each void pixel, the class whose nearest original boundary is closest wins. Crucially, only gap pixels ($L[p] = 0$) receive a new label; existing labels are never overwritten.

Step 3 — Re-vectorisation. The filled raster is polygonised, polygons smaller than $0.5\,\mathrm{m}^2$ are discarded, and the result is exported as a vector layer.

Raster gap filling reduces the polygon count by over 65 %, compared to at most 25 % with RAG. The proportion of small polygons drops from 40 % to around 20 %. The gain saturates quickly beyond a closing radius of 0.20 m, indicating that the remaining gaps are either wider than the structuring element or located between different classes. Table 3.3 shows the effect as the closing radius increases.

Max gap	Polygons	Reduction	$<5\,\mathrm{m}^2$	Median area
Baseline	1 973	–	40.0 %	$7.24\,\mathrm{m}^2$
0.15 m	688	65.1 %	20.6 %	$10.18\,\mathrm{m}^2$
0.20 m	670	66.0 %	20.0 %	$10.41\,\mathrm{m}^2$
0.25 m	659	66.6 %	19.9 %	$10.44\,\mathrm{m}^2$
0.50 m	629	68.1 %	19.9 %	$10.54\,\mathrm{m}^2$

Table 3.3: Effect of raster gap closing on polygon count and size distribution. The improvement plateaus beyond 0.20 m.

Both SMM and raster gap filling produce promising results ; although raster gap is greedier in term of fusion. Nevertheless, they rely on ground-truth class labels to guide the merging decisions. In the experiments above, labels were taken directly from the *Label_SAM_Disjoint* annotation layer, which is not available in a real deployment. The natural question is therefore: can we predict these labels automatically from the image data alone, without any manual annotation? The following section explores several approaches to this segment classification problem, from simple unsupervised clustering to supervised learning.



Figure 3.6: Effect of raster gap filling on `top_potsdam_2_10_RGB`. (a) Baseline *Label_SAM_Disjoint* (1 973 polygons). (b) After morphological closing with a 0.15 m radius (688 polygons). The large uniform regions are reconstructed and the micro-fragment clutter is substantially reduced.

3.5 Segment Classification

3.5.1 Objective and Feature Representations

SAM produces class-agnostic masks: each polygon has a geometry but no semantic label. A separate classification step is therefore needed to assign one of the six ISPRS categories (impervious surface, building, low vegetation, tree, car, clutter) to each segment, so that class-guided merging can operate without ground-truth annotations.

Four feature representations are evaluated in increasing order of richness.

RGB K-means. Each segment is described by its mean RGB value (3 dimensions). An oracle experiment, applying the same K-means on ground-truth segments instead of SAM segments, yields an identical purity of 0.43, confirming that the bottleneck is the feature space and not SAM's segmentation quality.

DINOv2 CLS. A square crop centred on the segment bounding box is passed through DINOv2 ViT-B/14 [12]; the CLS token (768 dimensions) serves as the descriptor. A tight 64 px crop, which forces the network to attend to the segment's own texture, outperforms a 224 px crop (accuracy 0.503 vs. 0.394), where surrounding context dilutes the signal.

DINOv2 multiscale. Three crops at 64, 128, and 256 px are embedded independently and concatenated into a 2 304-dimensional vector that combines fine-grained local texture with broader spatial context. Mean RGB statistics are also appended.

RF / MLP. Supervised classifiers are trained on a 4 622-dimensional feature vector that extends the DINOv2 multiscale embedding with geometric segment descriptors (area, perimeter, elongation) and mean RGB statistics.

3.5.2 Evaluation Protocol

Unsupervised. K-means is run without ground-truth labels; the resulting partition is evaluated a posteriori with three alignment-free metrics.

- **NMI** (Normalised Mutual Information, $[0, 1]$): shared information between the cluster and ground-truth partitions, normalised by their joint entropy.
- **ARI** (Adjusted Rand Index, $[-1, 1]$): concordant and discordant segment pairs, corrected for chance. Invariant to cluster count, making it the fairest metric for comparing k=6 and k=24.
- **Purity**: fraction of each cluster belonging to its majority class. Grows monotonically with k and therefore over-estimates discriminative quality when clusters are numerous.

For k=6, a Hungarian assignment maps clusters to semantic classes before reporting accuracy and F1. For k=24 (overclustering [7]), a greedy many-to-one assignment is used instead, allowing multiple sub-clusters to share the same class label.

Supervised. The Random Forest and MLP are evaluated with leave-one-tile-out cross-validation over the 19 Potsdam tiles: trained on 18 tiles, tested on the held-out tile, cycling through all 19 configurations. All reported metrics are averages over folds.

3.5.3 Global Results

Table 3.4 summarises all results on all 19 Potsdam tiles.

Model	Type	NMI	ARI	Purity	Acc.	Macro-F1	W-F1
RGB KMeans $k = 6$	Unsupervised	0.107	0.058	0.411	0.303	0.265	0.304
DINOv2 CLS $k = 6$	Unsupervised	0.320	0.232	0.563	0.502	0.508	0.531
DINOv2 CLS + RGB $k = 6$	Unsupervised	0.316	0.217	0.539	0.481	0.479	0.502
DINOv2 multiscale + RGB $k = 6$	Unsupervised	**0.369**	**0.310**	0.580	0.554	0.537	0.573
DINOv2 multiscale + RGB $k = 24$	Overclustering	0.352	0.157	**0.689**	**0.689**	**0.630**	**0.677**
Random Forest	Supervised LOO	0.499	0.530	0.743	0.743	0.686	–
MLP	Supervised LOO	**0.583**	**0.637**	**0.803**	**0.803**	**0.763**	–

Table 3.4: Global segment classification results averaged over the 19 Potsdam tiles. For unsupervised models, NMI, ARI and Purity are computed before cluster-to-class alignment, while Accuracy, Macro-F1 and Weighted-F1 are computed after a posteriori alignment. For $k = 6$, Hungarian one-to-one alignment is used. For $k = 24$, greedy many-to-one alignment is used. Random Forest and MLP are supervised baselines evaluated with leave-one-tile-out cross-validation; their NMI, ARI and Purity are computed from their predicted labels against the ground truth. For the supervised models, NMI, ARI and Purity are not clustering metrics in the strict sense, since RF and MLP directly predict semantic labels. They are reported only as complementary agreement measures between predicted labels and ground truth.

3.5.4 Interpretation

RGB is insufficient. K-means on mean RGB yields NMI = 0.107 and accuracy = 0.303. The six ISPRS classes span only two to three natural colour clusters on Potsdam imagery (bright surfaces, vegetation, shadow), so forcing k=6 imposes a prior that raw spectral colour cannot satisfy.

DINOv2 strongly improves. Replacing RGB with a single CLS embedding raises accuracy from 0.303 to 0.502 and NMI from 0.107 to 0.320. Semantic texture features capture structural patterns that colour alone misses.

Adding RGB to the CLS token does not help. Concatenating mean RGB to the CLS descriptor leaves NMI nearly unchanged (0.316) while reducing accuracy to 0.481. Colour information is already partially encoded by DINOv2; appending it explicitly perturbs the feature space without adding discriminative signal.

Multiscale DINOv2 with k=6 is the best honest unsupervised result. Combining three crop scales reaches NMI = 0.369 and ARI = **0.310**, the highest ARI among all unsupervised models. Since ARI corrects for chance and is invariant to cluster count, it provides the fairest basis for comparison across k values.

Overclustering to k=24 raises purity and accuracy, but the gain is partially optimistic. Purity climbs from 0.580 to 0.689 and post-alignment accuracy from 0.554 to 0.689, yet ARI drops from 0.310 to 0.157. This divergence — ARI $\downarrow$, Purity $\uparrow$ — is the mathematical fingerprint of overclustering: each sub-cluster becomes more internally homogeneous, so greedy many-to-one alignment captures intra-class appearance variability (e.g. red vs. grey rooftops, dense vs. sparse vegetation) rather than sharper inter-class boundaries.

The MLP defines the supervised upper bound. With the full 4 622-dimensional feature vector, the MLP achieves accuracy = 0.803 and macro-F1 = 0.763, confirming that the features carry sufficient semantic information when a small amount of labelled data is available. The gap with the best post-alignment unsupervised result accuracy (0.689) quantifies the headroom available to semi-supervised approaches. IIC [9], a label-free end-to-end alternative, did not surpass the DINOv2-based approaches on this dataset. A Graph Attention Network [18] exploiting segment spatial adjacency reaches macro-F1 = 0.695, slightly above the Random Forest (0.686) but

below the MLP, showing that local graph structure is complementary but insufficient to replace labelled training data.

3.5.5 Tile-level Analysis

Performance varies markedly across the 19 test tiles. The three best MLP tiles (`2_10`: macro-F1 = 0.796; `4_11`: 0.785; `2_13`: 0.779) share a balanced class distribution and spatially separated land-use units where SAM polygons tend to cover a single semantic class. The three worst tiles (`5_12`: F1 = 0.701; `5_11`: 0.742; `3_14`: 0.739) expose two distinct failure modes. Tiles `5_11` and `5_12` are dominated by continuous Haussmann-style apartment blocks (49–59 % of segments), whose flat rooftops are under-represented in the suburban training tiles, biasing predictions towards the building class. Tile `3_14` fails differently: a park-forest edge creates a spectral continuum between tree canopy and low vegetation, and SAM polygons spanning the transition carry mixed signals that neither class can claim. Across all models, buildings and cars are classified reliably everywhere (per-class F1 > 0.84 and > 0.93 respectively), while clutter remains the hardest class (F1 = 0.44–0.70) owing to its heterogeneous appearance and low frequency in training.

The same tile-level pattern holds across the unsupervised models, each exposing an additional model-specific failure mode: spectral ambiguity along infrastructure corridors (K-means), featureless open areas (DINOv2 k=6), and tree/grass spectral boundaries (MLP). Detailed per-model maps are provided in Appendix A.1.

3.5.6 Summary

DINOv2 multiscale embeddings are the core component of the classification pipeline: they break the RGB spectral bottleneck and produce descriptors rich enough for both unsupervised and supervised classifiers. Without any labels, multiscale K-means with k=6 achieves the best honest unsupervised result (NMI = 0.369, ARI = 0.310); overclustering to k=24 raises post-alignment accuracy to 0.689 by capturing intra-class appearance variability, at the cost of more diffuse cluster boundaries. The MLP supervised upper bound of 0.803 sets the target for future semi-supervised work. These predicted labels feed directly into the class-guided merging steps of Section 3.4, enabling SMM and raster gap filling to operate without ground-truth annotations.



Figure 3.7: MLP classification maps (left) and ground-truth ISPRS labels (right) for the best tile `2_10` (top, green border) and the worst tile `5_12` (bottom, red border). ISPRS colour scheme: grey = impervious surface, blue = building, light green = low vegetation, dark green = tree, yellow = car, red = clutter. Full results across all models are in Appendix A.1.

Chapter 4

Conclusion

This work examined the full processing pipeline for applying SAM to large aerial images, covering tiling strategy, mask merging, and segment classification. The central finding is that the order of operations matters: **classifying segments first, then merging per class**, is the correct design. Geometry-based merging without labels is not a viable path; class-guided merging with SMM and raster gap filling is.

What works. Overlapping tiling (20 % overlap) resolves boundary fragmentation for objects smaller than the tile size (30 m) by ensuring each object appears fully in at least one tile, a clear and robust improvement over non-overlapping grids. On classification, DINOv2 multiscale embeddings reliably break the RGB spectral bottleneck: without any labels, overclustering to k=24 reaches an accuracy of 0.689, and a supervised MLP trained on a small amount of labelled data reaches 0.803. In the oracle setting, once segments carry class labels, the class-guided merging pipeline becomes effective. The classification experiments then estimate how well these labels could be predicted automatically in a real deployment. Raster gap filling reduces polygon count by 65 % without introducing any cross-class fusions, and SMM achieves 49–63 % per-class reductions for the three smallest classes. Together, classification followed by class-guided merging produces a semantically coherent segmentation map, the combination of steps that geometry-only methods cannot achieve but with a a trade-off, because some fusion lead to small masks being merged with other masks belonging to the same class. But merging also leads to a reduction of small polygons within the same class, and overlapping grids tend to over-segment objects in general.

What does not work. Geometry-based merging (RAG, UCM) without class information is fundamentally limited: adjacent regions are fused on the basis of shared boundaries alone, inevitably merging semantically unrelated objects such as roads and buildings. The polygon count reduction they achieve (at most 25 %) comes at the cost of incorrect semantic groupings, making them unsuitable as standalone merging strategies. On the classification side, raw RGB K-means (accuracy 0.303) and adding RGB to DINOv2 CLS embeddings both fail to improve over DINOv2 alone, confirming that colour statistics carry no additional discriminative signal once semantic features are available. SMM does not scale to dominant classes due to ILP solver limitations, leaving those classes unmerged without the raster gap filling complement. Generalisation is uneven across scene types: the pipeline degrades on building-dominated tiles where a single class accounts for more than half of all segments (MLP macro-F1 down to 0.701 on `5_12`), and clutter remains the hardest class to classify reliably (F1 = 0.44–0.70).

Perspectives

The most impactful next step is to close the gap between the post-alignment unsupervised classification result (0.689) and the supervised bound (0.803). A semi-supervised approach, propagating a small number of user-labelled examples across the DINOv2 feature space, could substantially improve alignment without requiring a fully annotated training set, making the pipeline practical in real deployment scenarios where ground truth is scarce.

On the merging side, the remaining open problem is SMM scalability: the ILP solver times out for dominant classes (impervious surface, building, clutter), which together account for the majority of segments. Hierarchical or mini-batch ILP decomposition would extend class-based fusion to these large classes and complete the pipeline. Addressing class imbalance during training, through oversampling or class-balanced loss, would also reduce the performance gap between suburban and dense urban tiles.

Finally, the full pipeline should be validated beyond the Potsdam benchmark. Testing on additional sensors (near-infrared, LiDAR) and on datasets with different urban morphologies would assess whether the DINOv2 descriptors generalise to new acquisition conditions, and whether the classify-then-merge design holds as a general principle for class-agnostic segmentation models applied to remote sensing.

Skills acquired

This project provided hands-on experience with the full geospatial processing stack: vector and raster data manipulation with rasterio and geopandas, deep learning feature extraction with ViT-based models, combinatorial optimisation (ILP formulation and greedy approximations), and systematic experimental evaluation on a standard benchmark. Working with imperfect baselines on real data highlighted a key distinction that I will carry forward: what looks good visually and what is quantitatively correct are not the same thing, and building the tools to measure both is as important as the methods themselves.

Appendix A

Detailed Classification Results

A.1 Global Average Results

Table A.1 reports the average performance of all evaluated models. For unsupervised models, NMI, ARI and Purity are computed before cluster-to-class alignment. Accuracy, Macro-F1 and Weighted-F1 are computed after a posteriori alignment. For $k = 6$, Hungarian one-to-one alignment is used. For $k = 24$, greedy many-to-one alignment is used. The supervised Random Forest and MLP are evaluated with leave-one-tile-out cross-validation.

Model	NMI	ARI	Purity	Accuracy	Macro-F1	Weighted-F1
RGB KMeans $k = 6$	0.107	0.058	0.411	0.303	0.265	0.304
DINOv2 CLS $k = 6$	0.320	0.232	0.563	0.502	0.508	0.531
DINOv2 CLS + RGB $k = 6$	0.316	0.217	0.539	0.481	0.479	0.502
DINOv2 multiscale + RGB $k = 6$	**0.369**	**0.310**	0.580	0.554	0.537	0.573
DINOv2 multiscale + RGB $k = 24$	0.352	0.157	**0.689**	**0.689**	**0.630**	**0.677**
Random Forest	0.499	0.530	0.743	0.743	0.686	–
MLP	**0.583**	**0.637**	**0.803**	**0.803**	**0.763**	–

Table A.1: Global average classification and clustering results. Bold values indicate the best value among unsupervised models for the first five rows and the best supervised result for the last two rows.

A.2 Per-tile Macro-F1 for Unsupervised Models

Table A.2 reports the per-tile Macro-F1 after a posteriori cluster alignment.

Tile	RGB $k = 6$	DINOv2 $k = 6$	DINOv2+RGB $k = 6$	Multi $k = 6$	Multi $k = 24$
2_10	0.268	0.551	0.506	0.573	0.684
2_11	0.305	0.521	0.491	0.546	0.641
2_12	0.240	0.496	0.394	0.483	0.629
2_13	0.277	0.508	0.497	0.543	0.632
2_14	0.274	0.526	0.459	0.523	0.597
3_10	0.264	0.522	0.494	0.548	0.652
3_11	0.280	0.538	0.493	0.525	0.638
3_12	0.269	0.527	0.496	0.536	0.626
3_13	0.281	0.512	0.493	0.558	0.656
3_14	0.274	0.505	0.496	0.555	0.616
4_10	0.270	0.519	0.519	0.568	0.672
4_11	0.290	0.509	0.460	0.515	0.616
4_13	0.256	0.512	0.452	0.521	0.622
4_14	0.247	0.517	0.498	0.568	0.649
4_15	0.257	0.475	0.497	0.557	0.601
5_10	0.282	0.504	0.494	0.547	0.640
5_11	0.200	0.476	0.458	0.519	0.594
5_12	0.201	0.449	0.398	0.440	0.553
5_13	0.297	0.488	0.505	0.583	0.642

Table A.2: Per-tile Macro-F1 for all unsupervised models. For $k = 6$, Hungarian alignment is used. For $k = 24$, greedy many-to-one alignment is used.

A.3 Per-tile Leave-One-Out Results for Supervised Models

Table A.3 reports the leave-one-tile-out results for the supervised Random Forest and MLP classifiers.

Tile	RF Macro-F1	RF Accuracy	MLP Macro-F1	MLP Accuracy
2_10	0.714	0.698	0.795	0.817
2_11	0.734	0.715	0.774	0.763
2_12	0.719	0.735	0.748	0.764
2_13	0.700	0.718	0.779	0.788
2_14	0.732	0.768	0.778	0.811
3_10	0.688	0.748	0.777	0.803
3_11	0.659	0.726	0.778	0.817
3_12	0.695	0.752	0.756	0.798
3_13	0.682	0.756	0.768	0.812
3_14	0.671	0.734	0.739	0.782
4_10	0.729	0.749	0.771	0.785
4_11	0.710	0.755	0.785	0.816
4_13	0.670	0.764	0.761	0.825
4_14	0.687	0.735	0.765	0.793
4_15	0.660	0.729	0.768	0.815
5_10	0.709	0.736	0.750	0.765
5_11	0.642	0.779	0.742	0.840
5_12	0.594	0.792	0.701	0.848
5_13	0.636	0.725	0.763	0.810

Table A.3: Leave-one-tile-out supervised results for Random Forest and MLP.

A.4 Best and Worst Tiles by Model

Table A.4 reports the three best and three worst tiles for each model according to Macro-F1.

Model	Rank	Tile	Macro-F1
RGB KMeans $k = 6$	Best 1	2_11	0.305
RGB KMeans $k = 6$	Best 2	5_13	0.297
RGB KMeans $k = 6$	Best 3	4_11	0.290
RGB KMeans $k = 6$	Worst 1	5_11	0.200
RGB KMeans $k = 6$	Worst 2	5_12	0.201
RGB KMeans $k = 6$	Worst 3	2_12	0.240
DINOv2 CLS $k = 6$	Best 1	2_10	0.551
DINOv2 CLS $k = 6$	Best 2	3_11	0.538
DINOv2 CLS $k = 6$	Best 3	3_12	0.527
DINOv2 CLS $k = 6$	Worst 1	5_12	0.449
DINOv2 CLS $k = 6$	Worst 2	4_15	0.475
DINOv2 CLS $k = 6$	Worst 3	5_11	0.476
DINOv2 CLS + RGB $k = 6$	Best 1	4_10	0.519
DINOv2 CLS + RGB $k = 6$	Best 2	2_10	0.506
DINOv2 CLS + RGB $k = 6$	Best 3	5_13	0.505
DINOv2 CLS + RGB $k = 6$	Worst 1	2_12	0.394
DINOv2 CLS + RGB $k = 6$	Worst 2	5_12	0.398
DINOv2 CLS + RGB $k = 6$	Worst 3	4_13	0.452
DINOv2 multiscale + RGB $k = 6$	Best 1	5_13	0.583
DINOv2 multiscale + RGB $k = 6$	Best 2	2_10	0.573
DINOv2 multiscale + RGB $k = 6$	Best 3	4_14	0.568
DINOv2 multiscale + RGB $k = 6$	Worst 1	5_12	0.440
DINOv2 multiscale + RGB $k = 6$	Worst 2	2_12	0.483
DINOv2 multiscale + RGB $k = 6$	Worst 3	4_11	0.515
DINOv2 multiscale + RGB $k = 24$	Best 1	2_10	0.684
DINOv2 multiscale + RGB $k = 24$	Best 2	4_10	0.672
DINOv2 multiscale + RGB $k = 24$	Best 3	3_13	0.656
DINOv2 multiscale + RGB $k = 24$	Worst 1	5_12	0.553
DINOv2 multiscale + RGB $k = 24$	Worst 2	5_11	0.594
DINOv2 multiscale + RGB $k = 24$	Worst 3	2_14	0.597
Random Forest	Best 1	2_11	0.734
Random Forest	Best 2	2_14	0.732
Random Forest	Best 3	4_10	0.729
Random Forest	Worst 1	5_12	0.594
Random Forest	Worst 2	5_13	0.636
Random Forest	Worst 3	5_11	0.642
MLP	Best 1	2_10	0.795
MLP	Best 2	4_11	0.785
MLP	Best 3	2_13	0.779
MLP	Worst 1	5_12	0.701
MLP	Worst 2	3_14	0.739
MLP	Worst 3	5_11	0.742

Table A.4: Best and worst tiles for each evaluated model according to Macro-F1.



Figure A.1: RGB K-means (k=6) classification maps (left) and ground-truth ISPRS labels (right) for the best tile `2_11` (top, Macro-F1 = 0.305) and worst tile `5_11` (bottom, Macro-F1 = 0.200). Cluster collapse onto the dominant building class drives the low score on `5_11`. ISPRS colour scheme: grey = impervious surface, blue = building, light green = low vegetation, dark green = tree, yellow = car, red = clutter.



Figure A.2: DINOv2 CLS crop-64 + RGB (k=6) classification maps (left) and ground-truth ISPRS labels (right) for the best tile `2_10` (top, Macro-F1 = 0.506) and worst tile `5_12` (bottom, Macro-F1 = 0.398). Semantic DINOv2 features improve building/vegetation discrimination compared to pure RGB spectral clustering, yet dense Haussmann apartment blocks still collapse into a single dominant cluster.

Appendix B

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